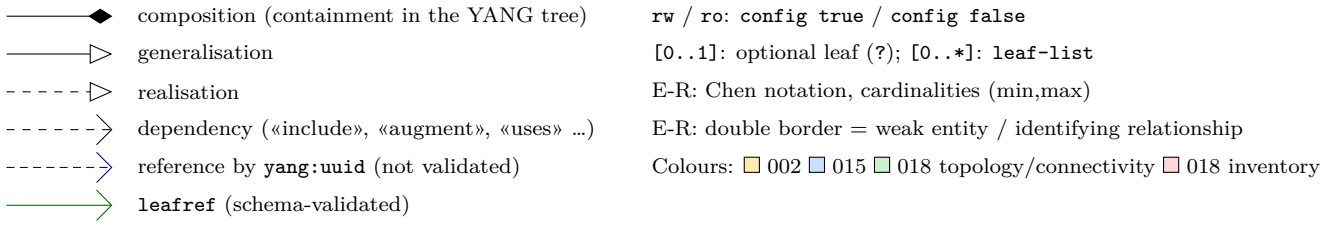
UML and E-R diagrams
ETSI GS QKD 002, 015, 018

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This collection describes the three ETSI documents with UML (use case and class) and E-R diagrams. For QKD 002, which defines no data model, the diagrams are a *conceptual model* derived from clauses 4, 6 and 7. For QKD 015 and 018 the diagrams follow the structure of the YANG modules: every `container` or `list` becomes a class, every `leaf` an attribute, and the tree hierarchy becomes composition.

Sources. QKD 015: modules `etsi-qkd-sdn-node` and `etsi-qkd-node-types`, revision 2022-01-30 (V2.1.1), checked attribute by attribute with `pyang -f tree`. QKD 018: modules `etsi-qkd-sdn-topology`, `-inventory`, `-connectivity` (revision 2022-02-22, ETSI Forge `gs018-orch-int-sdn`), also checked with `pyang`. *Italics* mark annotations (units, comments), not YANG names.

Notation



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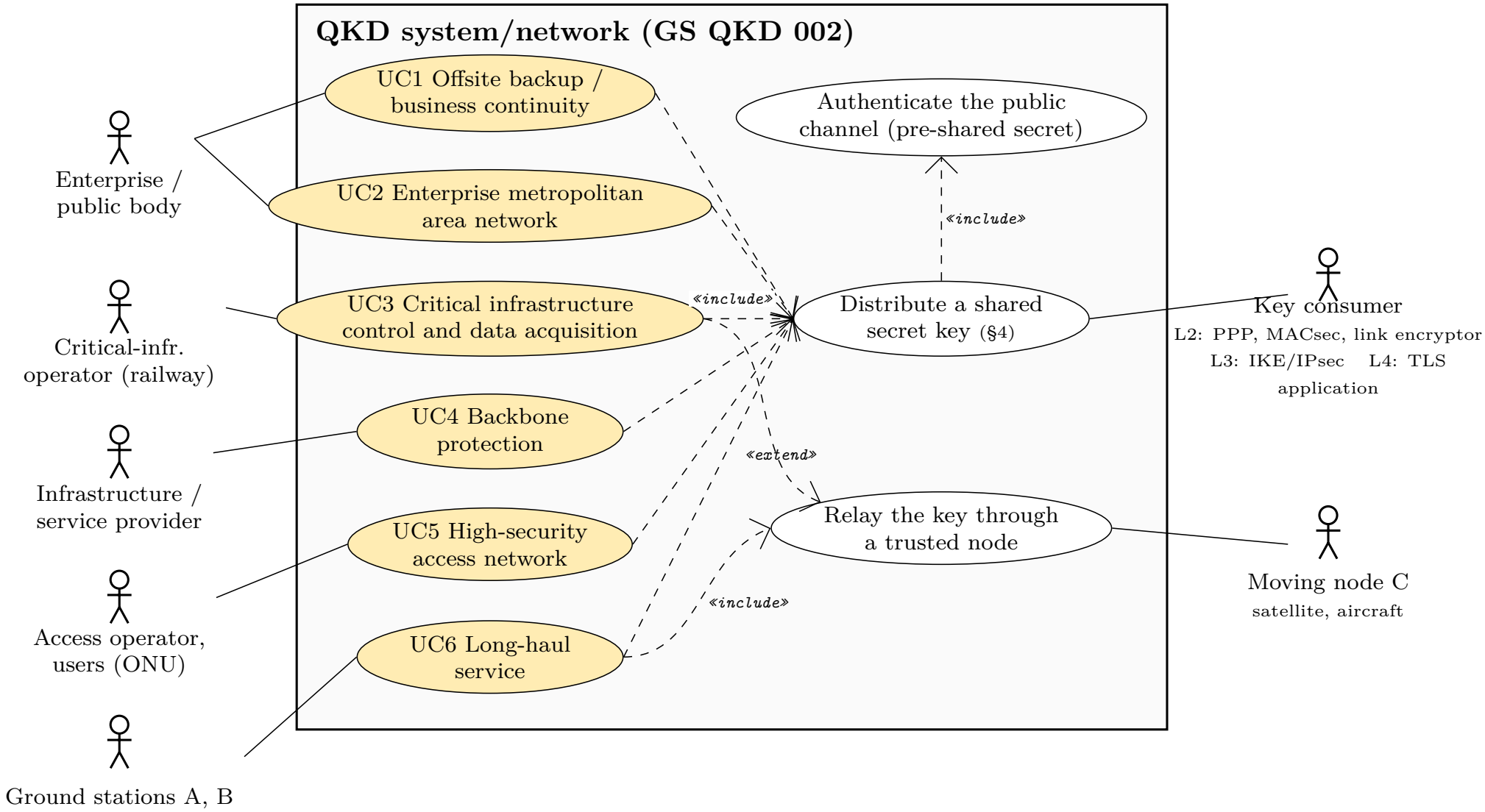


Figure 1: GS QKD 002 – use case diagram: the six cases of clause 7, all relying on the distribution of a shared secret key (§4), which requires an authenticated public channel; key relay through a trusted node is part of UC6 and one of the architectures of UC3.

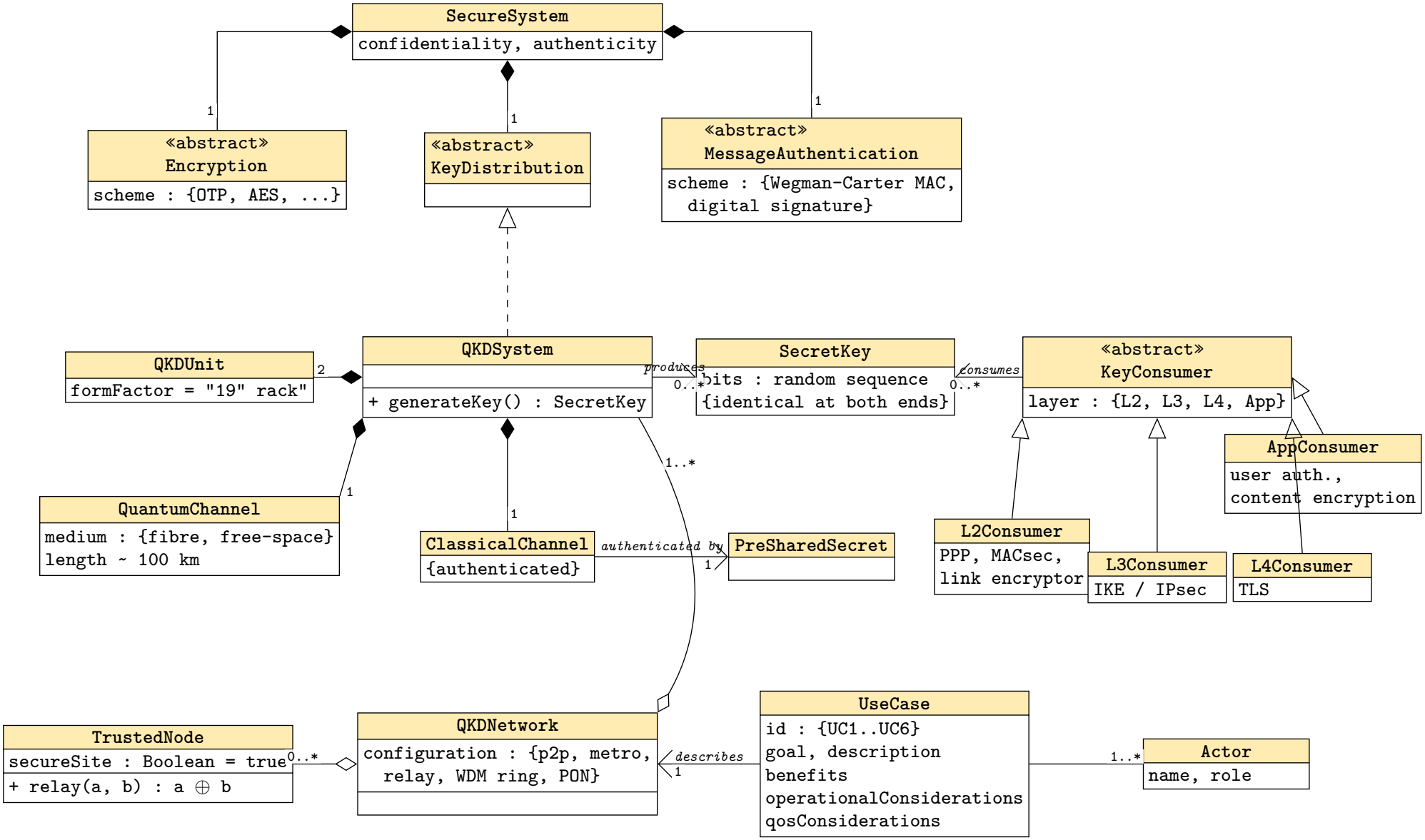


Figure 2: GS QKD 002 – conceptual class model: QKD realises only the key-distribution primitive (§4); the key is consumed at layers L2–L7 (§6); the network with trusted nodes and the use cases come from clause 7.

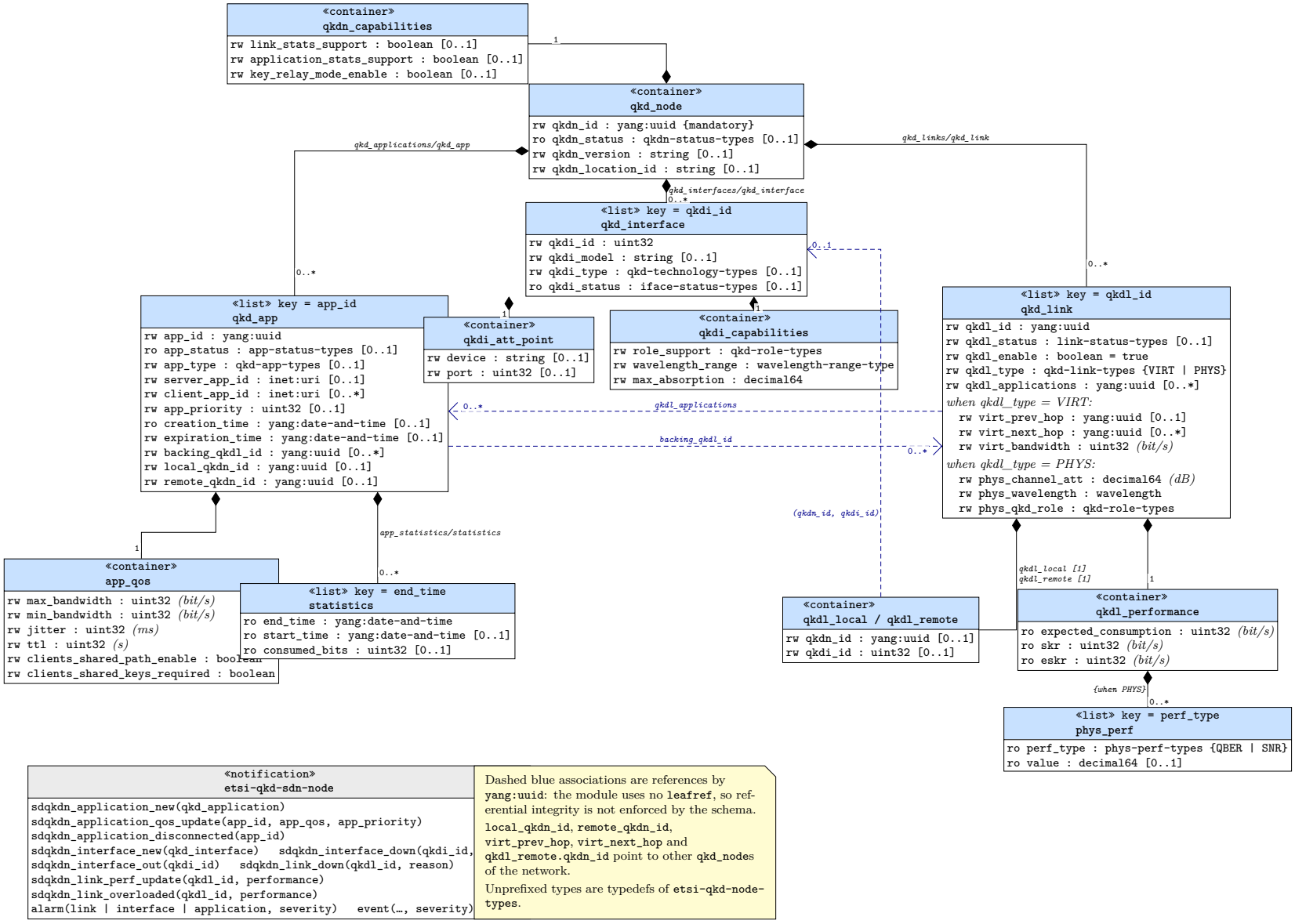


Figure 3: GS QKD 015 – class diagram of the `etsi-qkd-sdn-node` module: the `qkd_node` root holds capabilities, applications, interfaces and key association links, plus the module's notifications.

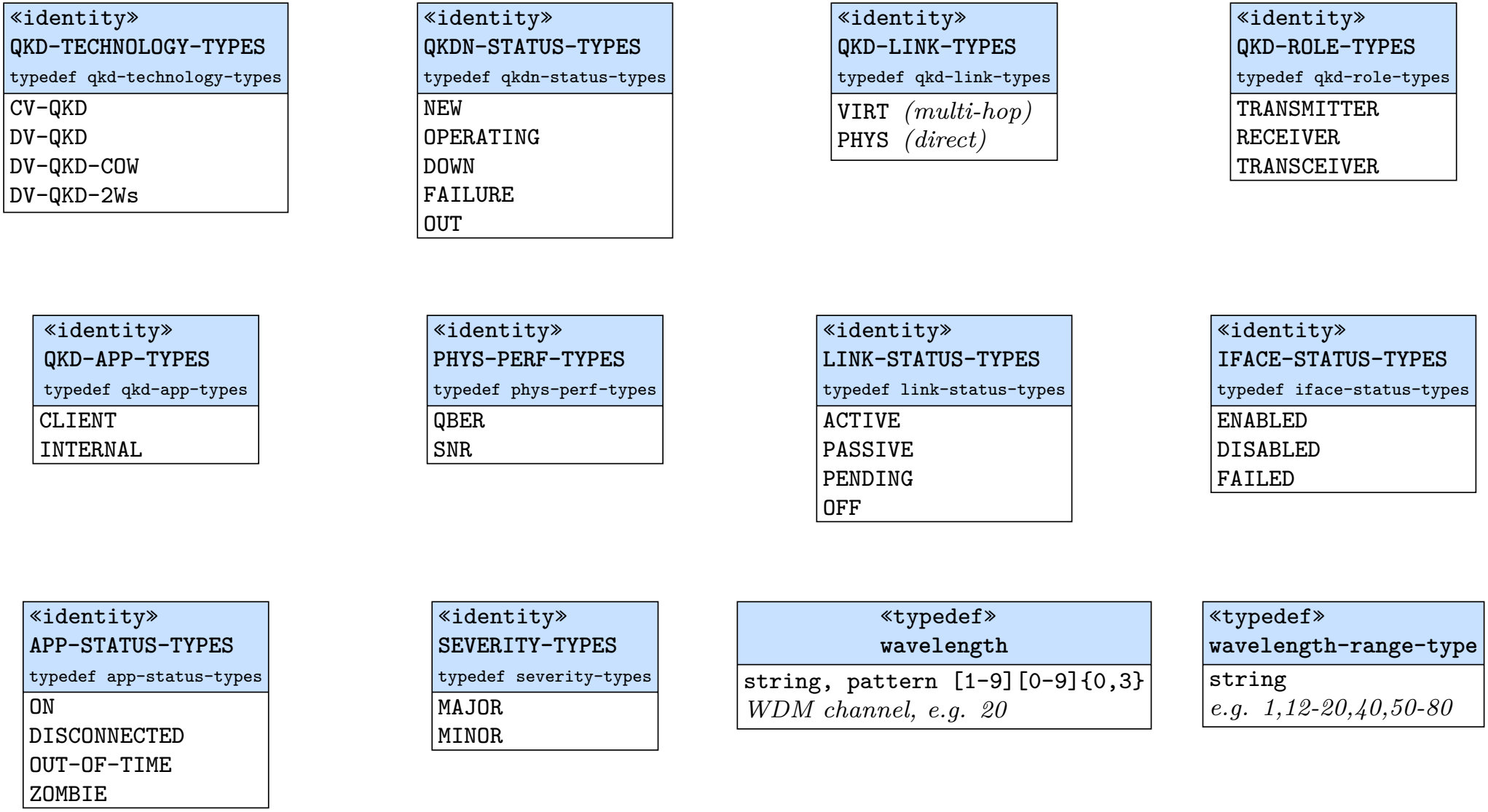


Figure 4: GS QKD 015 – identities and typedefs of the `etsi-qkd-node-types` module, also imported by QKD 018.

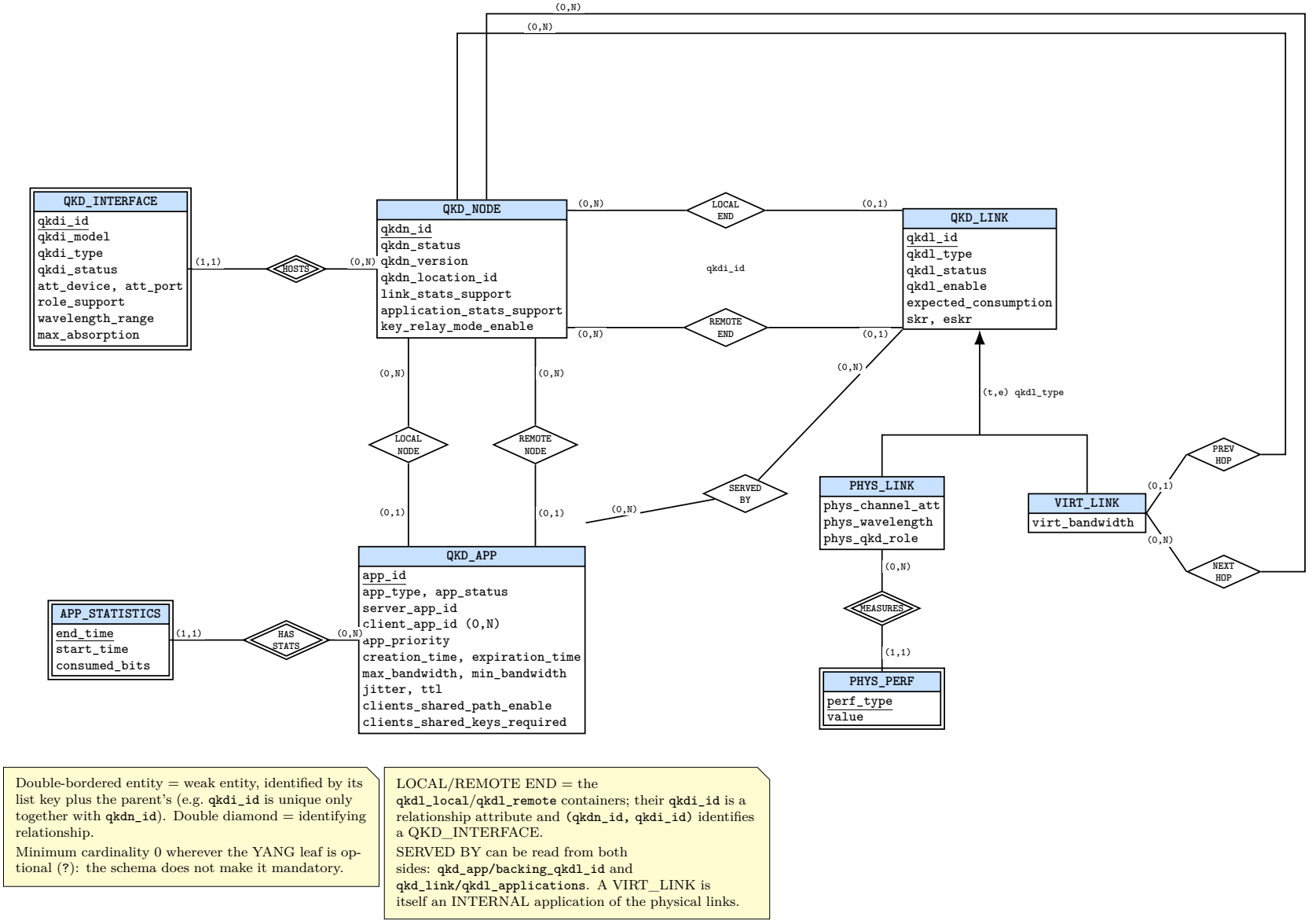


Figure 5: GS QKD 015 – E-R schema of the data the controller collects from the nodes.

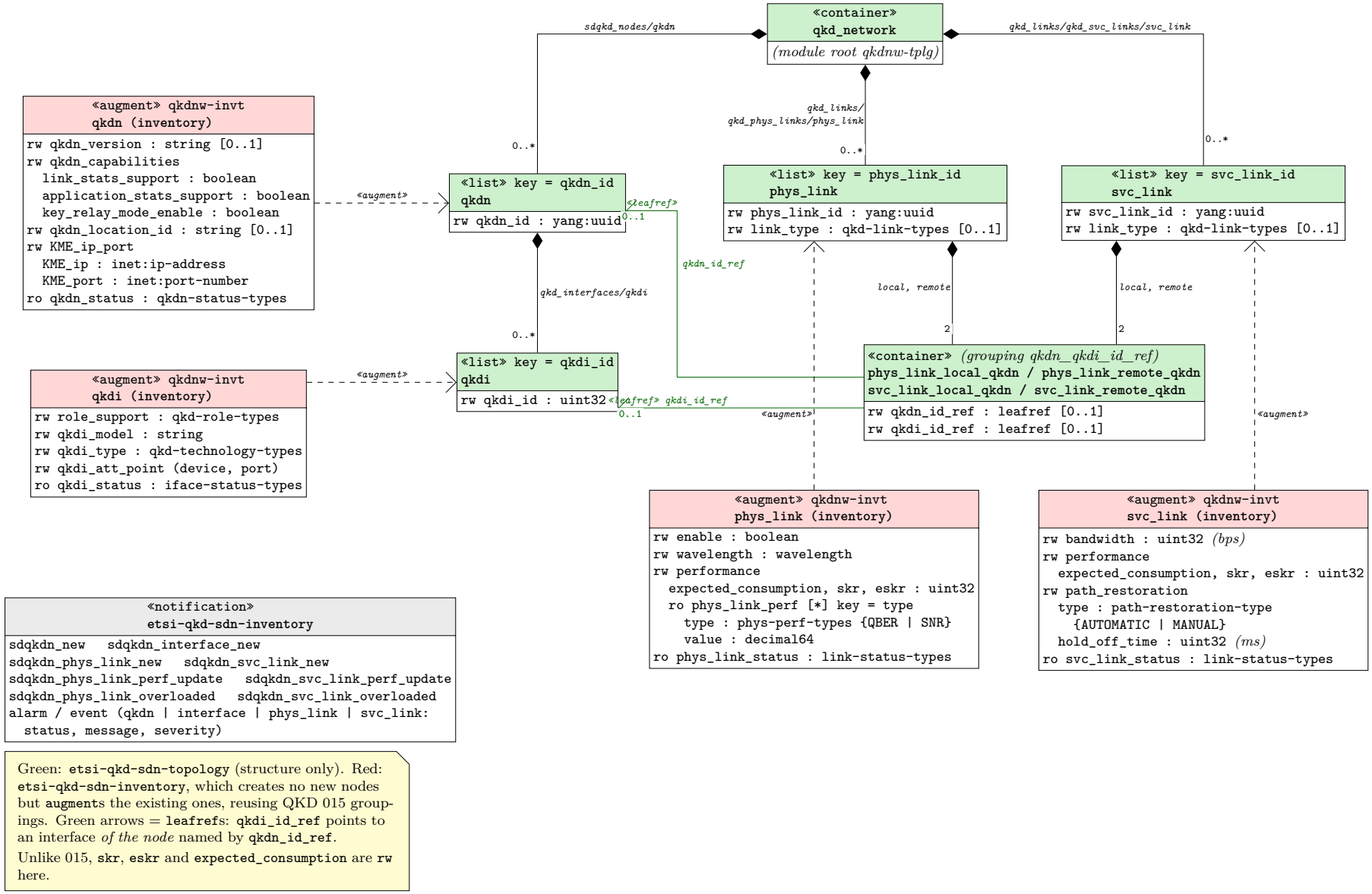


Figure 6: GS QKD 018 – class diagram of `etsi-qkd-sdn-topology` and of the `augments` of `etsi-qkd-sdn-inventory`.

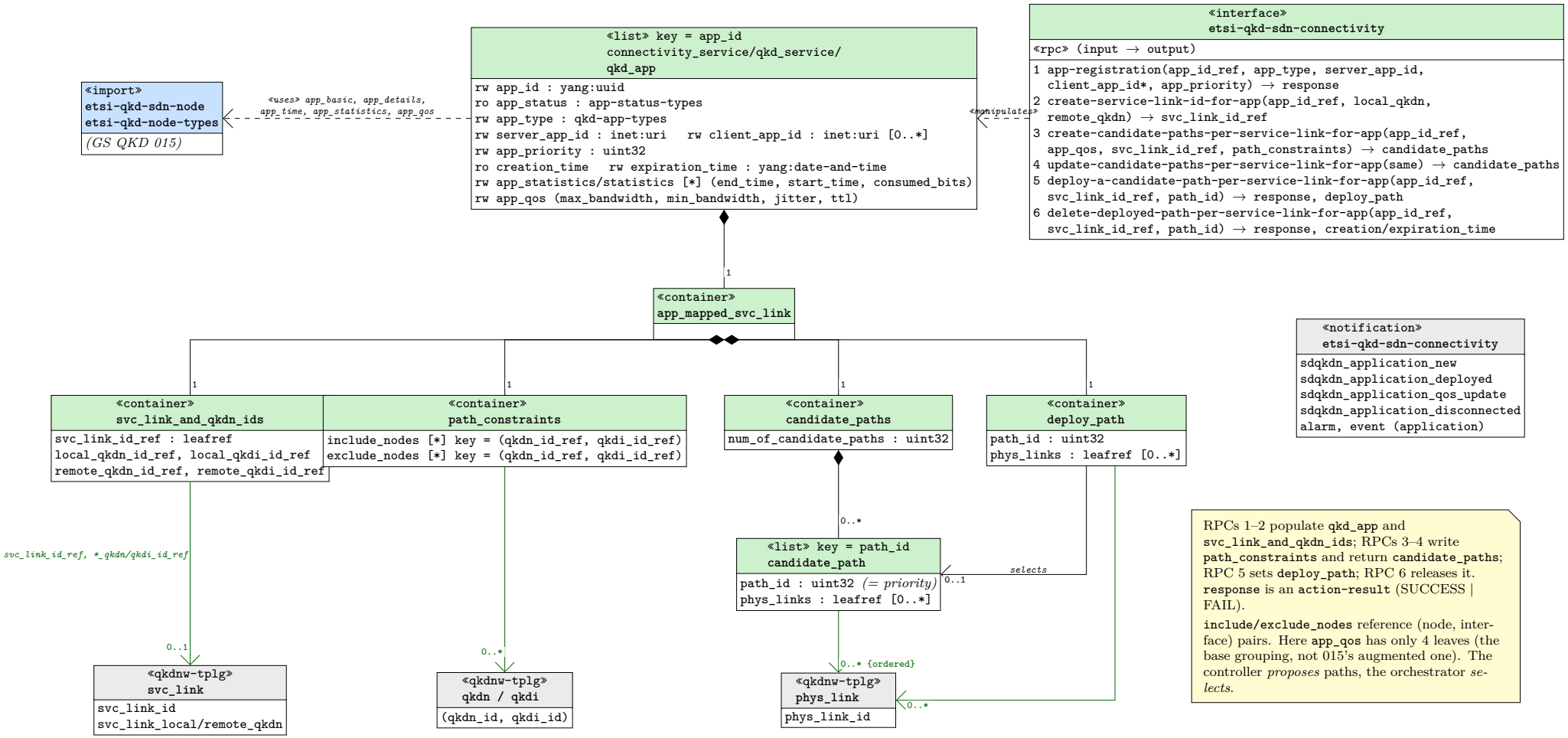


Figure 7: GS QKD 018 – class diagram of `etsi-qkd-sdn-connectivity`: application, mapped service link, constraints, candidate paths and chosen path, with the RPCs that manipulate them.

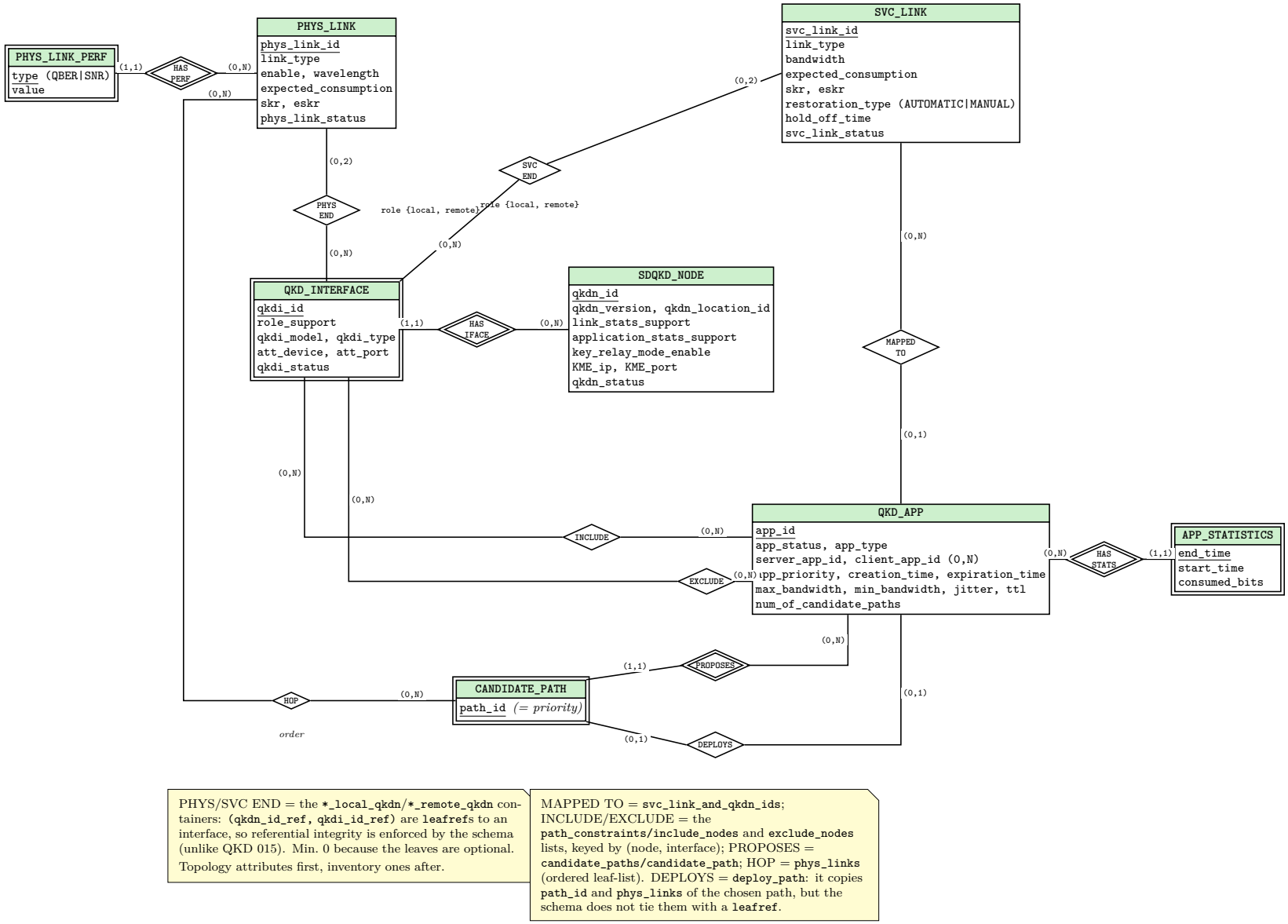


Figure 8: GS QKD 018 – E-R schema of the orchestrator's view of one QKD domain (the three modules together).

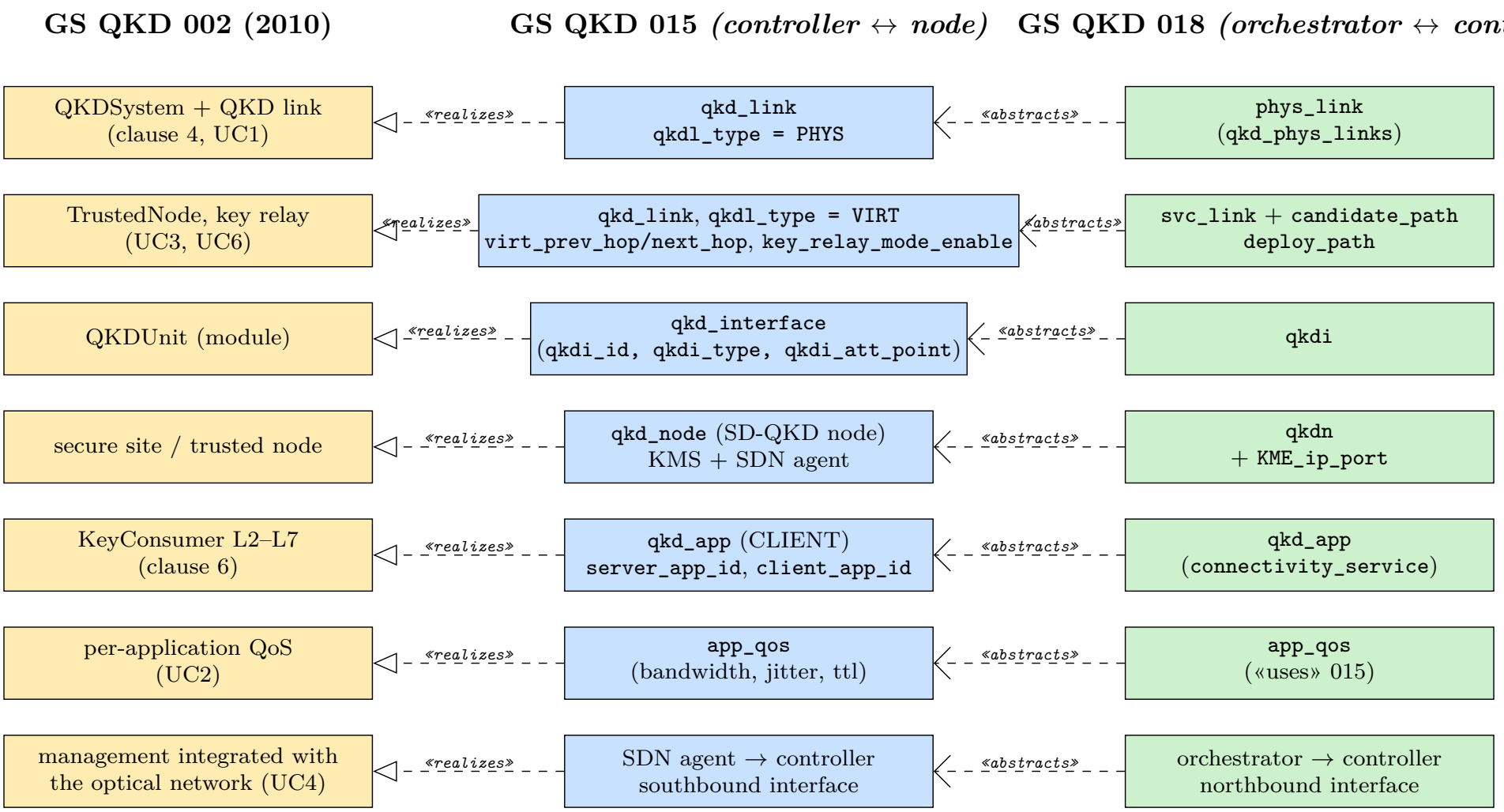


Figure 9: Traceability: how the QKD 002 concepts are realised by the QKD 015 node model and abstracted by the QKD 018 orchestration model.